



80m / 140m / 200m Dual Laser Calibration

Laser Rangefinder Module

Key Properties

- Measurement Range: 0.05m ~ 80/140/200m (indoor)
- Frequency: 1~20 Hz
- Accuracy: $\pm 1\text{mm}$ @ 40m
 $\pm 2\text{mm}$ @ 80m
 $\pm 7\text{mm}$ @ 140m
 $\pm 15\text{mm}$ @ 200m
- Wavelength: 650 nm
- Laser Safety: Class 2 / Class3R
- Communication Interface: TTL (UART)
- Dual Laser Calibration



Function

- | | |
|--------------------------|-------------------|
| - Single Measurement | - Address Setting |
| - Continuous Measurement | - Offset Setting |
| - Fast/Slow/Auto Mode | - Laser ON/OFF |

Introduction

The LRFX0M20PD is a high-precision laser rangefinder module with dual-laser calibration, eliminating environmental and electronic errors to deliver superior stability and accuracy over single-laser modules, while maintaining absolute accuracy under varying conditions.

The RFX0M20PD is designed for indoor measurements of static targets. This module prioritizes providing accurate measurement results, so its frequency will decrease or may not provide results when measuring moving targets or objects with lower reflectivity. In such cases, you may consider [Dynamic Dual-Laser Calibration Laser Rangefinder Module](#)

Note: When using the LRF140M20PD or LRF200M20PD for targets over 100 meters away, if the target's reflectivity is below 70%, a reflective film should be applied to improve reflectivity.



1. Specifications

Technical specifications			
Model Name	LRF80M20PD	LRF140M20PD	LRF200M20PD
Measurement Range	0.05m ~ 80m / 140m / 200m (indoor)		
Frequency	1~20Hz		
Accuracy	±1mm (at 40m) ±2mm (at 80m) ±7mm (at 140m) ±15mm (at 200m)		
Wavelength	650nm		
Laser Safety	Class2	Class3R	Class3R
Communication			
Communication Interface	3.3V TTL (UART)		
Baud Rate	9600 bps		
Interface Connector	8 Position 0.5 mm FFC & FPC Connectors		
Electrical			
Input Voltage	3.3~4.2V DC		
Power Consumption	120 mA		
Mechanical			
Dimensions	46.5 x 37 x 17.4mm		
Weight	22±1g		
Protection	IP40		
Environmental			
Operating Temperature	-10°C~50°C		
Storage Temperature	-20°C~80°C		

Note:

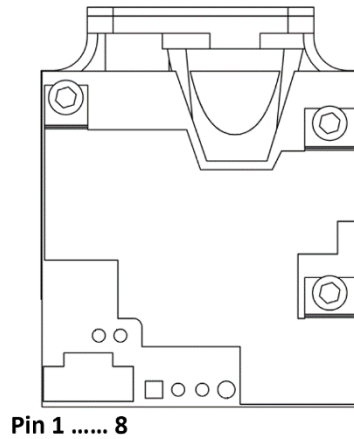
1. Different measurement targets and environments can cause a reduction in measuring range or a large measurement error due to factors such as excessively high ambient light intensity, extremely high or low ambient temperatures, weak or strong reflection from the target, or rough and uneven target surfaces.
2. The measurement range and accuracy are based on indoor conditions with targets having 70% reflectivity.
3. When measuring targets over 100 m with the LRF140M20PD or LRF200M20PD, ensure the target's reflectivity is above 70%. If the reflectivity is lower, a reflective film should be used to enhance it.



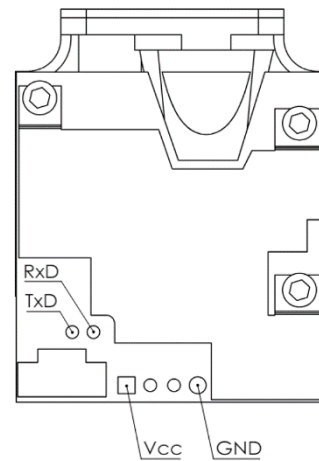
2. Pin Assignment

LRFX0M20PD provides FPC connector and soldering points.

1	Vcc
2	Vcc
3	GND
4	GND
5	Vcc
6	Vcc
7	TxD
8	RxD

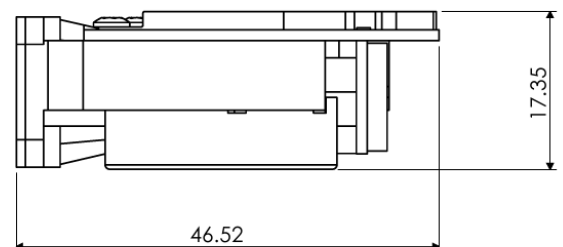
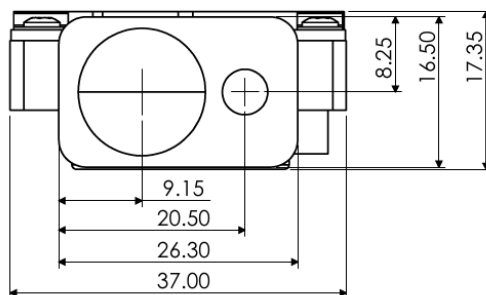
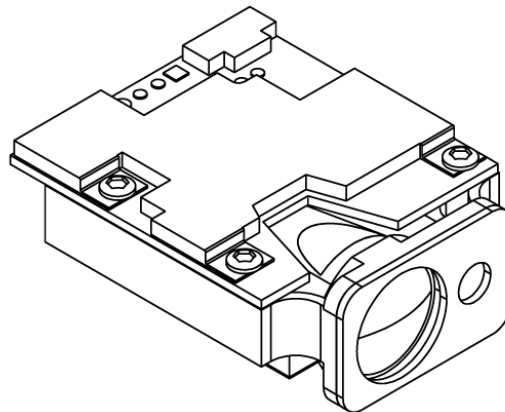
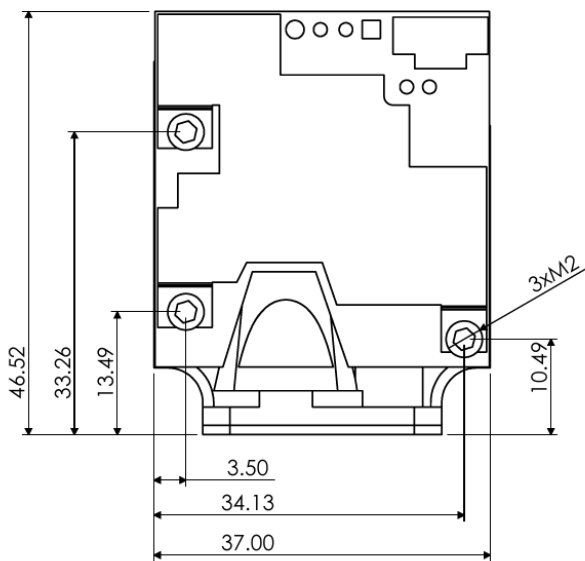


FPC Connector



Soldering Points

3. Dimensions





4. Communication

4.1 UART configuration parameters:

8N1 with default 9600 baud rate, byte data are expressed in hexadecimal.

Data bits: 8

Parity bits: None

Stop bits: 1

4.2 Command List

Function	Chapter
Single Measurement	4.2.1
Continuous Measurement	4.2.2
Stop Measurement	4.2.3
Laser ON/OFF	4.2.4
Set Address	4.2.5
Set Frequency	4.2.6
Set Offset	4.2.7
Set Resolution	4.2.8
Set Interval Time	4.2.9
Measure Upon Power-On Enable/Disable	4.2.10
Single Measurement (broadcast)	4.2.11
Read Buffer	4.2.12

***ADD** is the address of the module, with a default address of 0x80. The broadcast address is 0xFA.

***CHK** is the checksum, sum up all the preceding bytes and take the complement.

Example: 0xFA, 0x06, 0x04 CHK

CHK=0xFA + 0x06 + 0x04 = 0x04(carry ignored)

The complement of 0x04 is 0xFC, CHK is 0xFC.

4.2.1 Single Measurement

Send	ADD 06 02 CHK Example: 80 06 02 78
Receive	ADD 06 82 3X 3X 3X 2E 3X 3X 3X CHK (Under the resolution setting of 1mm) ADD 06 82 3X 3X 3X 2E 3X 3X 3X 3X CHK (Under the resolution setting of 0.1mm)

*The **bold text** represents the measurement result data, which is in ASCII format. For example, the measurement result "123.456 meters" is displayed as "31 32 33 2E 34 35 36" in ASCII format.

* Please refer to chapter 4.2.8 for the configuration of 1mm/0.1mm resolution setting.

* When measurement fails, return "ADD 06 82 'E' 'R' 'R' '-' '-' 3X 3X CHK" (in 1mm resolution)

* When measurement fails, return "ADD 06 82 'E' 'R' 'R' '-' '-' '-' 3X 3X CHK" (in 0.1mm resolution)



4.2.2 Continuous Measurement

Send	ADD 06 03 CHK Example: 80 06 03 77
Receive	ADD 06 83 3X 3X 3X 2E 3X 3X 3X CHK (Under the resolution setting of 1mm) ADD 06 83 3X 3X 3X 2E 3X 3X 3X 3X CHK (Under the resolution setting of 0.1mm)

*The **bold text** represents the measurement result data, which is in ASCII format. For example, the measurement result "123.456 meters" is displayed as "31 32 33 2E 34 35 36" in ASCII format.

* Please refer to chapter 4.2.8 for the configuration of 1mm/0.1mm resolution setting.

* When measurement fails, return "ADD 06 83 'E' 'R' 'R' '-' '-' 3X 3X CHK" (in 1mm resolution)

* When measurement fails, return "ADD 06 83 'E' 'R' 'R' '-' '-' '-' 3X 3X CHK" (in 0.1mm resolution)

4.2.3 Stop Measurement

Send	ADD 04 02 CHK Example: 80 04 02 7A
Receive	ADD 04 82 CHK

4.2.4 Laser ON/OFF

Send	ADD 06 05 01 CHK (Laser ON) ADD 06 05 00 CHK (Laser OFF) Example: 80 06 05 01 74 (Laser ON)
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4.2.5 Set Address

Send	ADD_old 04 01 ADD_new CHK Example: FA 04 01 80 81 (set address as 0x80)
Receive	ADD 04 81 CHK (Address setting successful) ADD 84 81 02 CHK (Address setting failed)

*ADD_old is the original address, and **ADD_new** is the new address to be set.

*The address setting range is 0x00 ~ 0xFF.

*The default address is 0x80.

*The broadcast address is 0xFA.

4.2.6 Set Frequency

Send	ADD 04 0A 00 CHK (3Hz) ADD 04 0A 05 CHK (5Hz) ADD 04 0A 0A CHK (10Hz) ADD 04 0A 14 CHK (20Hz) Example: FA 04 0A 00 F8 (set frequency as 3Hz)
Receive	ADD 04 8A CHK (Frequency setting successful) ADD 84 8A 01 CHK (Frequency setting failed)



4.2.7 Set Offset

Send	ADD 04 06 SIGN DIST CHK Example: FA 04 06 2D 01 CE (set offset as -1)
Receive	ADD 04 8B CHK (Offset setting successful) ADD 84 8B 01 CHK (Offset setting failed)

***SIGN** is the positive or negative offset value, where 0x2D is negative and 0x2B is positive.

***DIST** is the offset to be set.

4.2.8 Set Resolution

Send	ADD 04 0C 01 CHK (1mm) ADD 04 0C 02 CHK (0.1mm) Example: FA 04 0C 01 F5 (set resolution as 1mm)
Receive	ADD 04 8C CHK (Resolution setting successful) ADD 84 8C 01 CHK (Resolution setting failed)

4.2.9 Set Interval Time

Send	ADD 04 05 INTE CHK Example: FA 04 05 01 FC (set interval as 1sec)
Receive	ADD 04 85 CHK (Interval time setting successful) ADD 84 85 01 CHK (Interval time setting failed)

***INTE** represents the interval time (sec) to be set.

* Interval Time is the time interval between each measurements during continuous measurement.

4.2.10 Measure Upon Power-On Enable/Disable

Send	ADD 04 0D 00 CHK (Disable) ADD 04 0D 01 CHK (Enable) Example: FA 04 0D 01 F4 (Enable Measure Upon Power-On)
Receive	ADD 04 8D CHK (Setting successful) ADD 84 8D 01 CHK (Setting failed)

*When Enable status is set, the module will start continuous measurement immediately after power-on.

4.2.11 Single Measurement (broadcast)

Send	FA 06 06 FA
Receive	* No response from module. For measurement results please use Read Buffer command. Please refer to Section 4.2.12 for details.

4.2.12 Read Buffer

Send	ADD 06 07 CHK
Receive	*Please refer to Section 4.2.1, "Receive" part, for Single Measurement.



5. Mounting

The installation method for LRFX0M20PD recommends replacing the three black screws at the bottom of the rangefinder module with M2 copper pillars. Please refer to the following images:

